

Carlos Manuel Martínez Vázquez

Computer Science Student @ Monterrey Institute of Technology

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SUMMARY

Computer Science student and Quantitative Researcher with a strong foundation in statistics, optimization, and algorithmic modeling. Experienced in ML-based alpha signal extraction and robust cross-sector allocation using advanced econometrics. interested in quantitative modeling, time-series forecasting, and robust deep learning architectures.

EDUCATION

Monterrey Institute of Technology

B.S. in Computer Science and Technology (GPA: 3.7/4.0)

Honors: Academic Excellence Scholarship.

Querétaro, México

Aug 2023 – June 2027

EXPERIENCE

Quantitative Researcher Intern

Moreton Capital Partners

Mar 2026 – Present

Mexico City, México

Engineering systematic research workflows for multi-horizon signal blending within commodity markets.

Developing regime-aware portfolio optimization frameworks and risk-parity allocation models to navigate volatility and architecting robust data engineering pipelines to ingest and process high-dimensional financial datasets for alpha extraction.

Lead Researcher | *Hybrid Asset Pricing Architecture*

Working Paper

Dec 2025 – Present

Engineering a novel data-cleaning and engineering architecture utilizing a Multilayer Perceptron (MLP) to filter stochastic noise from financial text prior to processing by LLM or BERT-based systems.

Designing high-fidelity preprocessing layers to isolate alpha signals in high-volatility asset classes and developing a hybrid Bi-LSTM architecture to integrate filtered sentiment scores into predictive pricing models.

RESEARCH & PROJECTS

Hierarchical Commodity Portfolio Construction

Mar 2026

Developed a two-stage backtesting engine for 10 commodity sectors over 162 weekly periods with SLSQP min variance optimization algorithm that achieved max 7.7% annual volatility.

Enforced KKT optimality to maintain stable weights and model transactions and achieved a 4.02 Adjusted Sharpe (Andrew W Lo 2002) and 2.1 Calmar Ratio by implementing Ledoit-Wolf shrinkage to regularize covariance matrices.

OrbitGrow: Martian Plant Health Monitoring | *Syngenta @ START Hack*

Mar 2026

Engineered a real-time computer vision system to monitor crop health in Martian greenhouses, utilizing AWS SageMaker for autonomous stress detection and awarded Top 5 Finalist out of 30 teams in the Syngenta Challenge at START Hack St. Gallen, Switzerland..

Developed a multi-modal data pipeline to process environmental telemetry and spectral imagery, enabling automated nutrient deficiency diagnosis in extraterrestrial environments.

AquaHub: Predictive Drought Modeling & Crowdsourcing

Feb 2026

Engineered a predictive ML model for drought detection in CDMX, processing 960+ training samples to identify water scarcity hotspots and validated a 5-day sliding window architecture, achieving 70–85% accuracy in preliminary drought event detection.

Designed a validation pipeline to cross-reference crowdsourced reports against meteorological norms, covering a demographic of 17.7M residents.

Bloomly: Global Bloom Detection System | *@ NASA Space Apps Hack*

Oct 2025

Engineered a LightGBM-based predictive model leveraging multi-spectral satellite and meteorological data and conducted feature selection and dimensionality reduction across 44 ecological indicators to improve classification precision and evaluated model performance via AUC and F1 metrics under temporal cross-validation for predictive stability.

Deployed Flask APIs with interactive 3D data visualization (Three.js) for spatial analysis of global bloom patterns.

TECHNICAL SKILLS

Languages & Quant: Python, C++, R, SQL, JavaScript, scikit-learn, LightGBM, pandas, NumPy, PyTorch.

Dev & Cloud: Snowflake, React, Next.js, Node.js, Flask, PostgreSQL, AWS, Git. **Spoken:** Spanish (Native), English (C1).